

Risk Adjustment Coding Opportunity Identification Platform

Technical Design & Methodology Notes

Compiled from project notebook · Medicare Advantage / Claims / EHR / Risk Adjustment Pipeline

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1. Data Tables

The platform combines longitudinal Medicare Advantage data from multiple operational systems. Each table contributes a different view of the member, provider, or encounter.

Table	Description
Member Eligibility	Enrollment, coverage dates, plan type, RAF score, dual eligibility
Medical Claims	ICD-10 diagnoses, CPT procedures, paid amount, allowed amount, provider specialty
EHR Notes	Assessment, plan, SOAP notes, physician documentation
Problem List	Persistent chronic conditions maintained by providers
Lab Results	HbA1c, eGFR, creatinine, BNP, LDL and other clinical indicators
Medication History	Prescription fills and therapeutic classes
Provider Master	Specialty, organization, coding audit history, coding accuracy
Historical Diagnoses	Prior ICD-10 and HCC history across previous encounters

Member Demographic Details

Field	Description
Member_ID	Unique member identifier
Age	Member age
Gender	Male / Female / Other
State	State of residence
Plan Type	Medicare Advantage product
RAF Score	Risk Adjustment Factor used by CMS
Dual Eligible	Medicare + Medicaid indicator

2. Problem Statement

Goal: Identify encounters where a clinically supported chronic condition may be missing from submitted diagnosis codes, creating a potential HCC coding opportunity.

Qualified Population Criteria

- Medicare Advantage members with active enrollment.
- Continuous eligibility during the observation window.
- At least one professional or facility claim in the last 12 months.
- At least one historical diagnosis, medication, or lab record available.

Business Question

Which encounters are likely to contain an undocumented HCC diagnosis that should be reviewed by a certified coder?

3. Business Impact

- **Improve Medicare Advantage risk adjustment accuracy.**
- **Reduce revenue leakage from undocumented chronic conditions.**
- **Prioritize coder workload toward high-value encounters.**
- **Increase transparency through explainable recommendations.**

A payer with millions of Medicare Advantage members can recover significant RAF revenue by improving HCC capture accuracy while maintaining coding compliance.

4. Data Collection & Model Design

Scale

- ~5 – 10 million Medicare Advantage members.
- 24 months of historical claims and clinical data.
- Encounter-level scoring.
- Separate opportunity models for major HCC categories (CHF, CKD, COPD, Diabetes, etc.).

Label Engineering

Target = 1 when retrospective review, audit findings, or a subsequent validated encounter confirms that an HCC diagnosis was clinically present but not coded on the current encounter. Otherwise Target = 0.

5. Data Pipeline Steps

Step 1 — Eligible Encounter Selection

Filter active Medicare Advantage encounters.

Step 2 — Join Claims

Attach ICD-10, CPT, paid amount, provider specialty.

Step 3 — Join EHR Notes

Extract clinical evidence from physician documentation.

Step 4 — Join Labs

Attach latest lab values and abnormality indicators.

Step 5 — Join Medications

Create therapeutic-class features.

Step 6 — Join Historical Diagnoses

Compute diagnosis persistence and prior HCC indicators.

Step 7 — Data Cleaning

Impute missing values and cap extreme outliers.

Step 8 — Encoding

One-hot encode categorical variables.

6. Feature Selection

- Drop zero-variance columns.
- Remove highly correlated features.
- Apply Recursive Feature Elimination (RFE).
- Reduce features from ~15k – 20k engineered variables to ~1k high-value predictors.

Examples of high-value features

Feature Group	Example
Diagnosis Persistence	CHF documented in 2 of last 3 visits
Medication Consistency	Loop diuretic + beta blocker + ACE inhibitor
Lab Indicators	Elevated BNP, low eGFR
Provider Behavior	Historical coding opportunity rate
Utilization	Multiple cardiology visits in last 6 months
Problem List Overlap	CKD present on problem list but absent on claim

7. Modeling Approach

Primary Model

CatBoost / XGBoost binary classifier trained on encounter-level features.

Output

- Missing HCC probability (0 – 1)
- Risk category: Low / Medium / High
- Top contributing features

Example

Encounter ID: 90887766
Predicted Missing HCC Probability: 0.92
Risk Category: High
Suspected HCC: Congestive Heart Failure

8. Model Evaluation

Metric	Purpose
AUROC	Ranking quality
Precision	Coder workload efficiency
Recall	Missed opportunity reduction
F1 Score	Balance between precision and recall
Calibration	Probability reliability

Operational Thresholding

Thresholds are selected jointly with coding operations. A lower threshold increases recall but creates more coder reviews; a higher threshold improves precision but may miss opportunities.

9. Post-Modeling Analysis

- Provider-wise opportunity rate
- Specialty-wise coding patterns
- Region-wise variation
- HCC-wise opportunity distribution
- Financial impact by provider group
- False positive review analysis

Analysis Area	Example Finding
Provider Pattern	Cardiology Group A shows 18% higher CHF opportunity rate
Specialty Trend	Nephrology has highest CKD documentation gap
Regional Trend	Texas region shows rising CKD opportunities
Financial Impact	Estimated RAF opportunity concentrated in CHF and CKD

10. Bias & Fairness Checks

Evaluate fairness across age groups, gender, dual-eligible status, and geographic regions.

Metric	Interpretation
Demographic Parity	Positive prediction rate consistency
Equal Opportunity	True positive rate consistency
Equalized Odds	TPR and FPR consistency
Disparate Impact	Ratio-based fairness assessment

11. SHAP Explainability

Example encounter explanation

Feature	Impact
Prior CHF diagnosis	+0.31
Lasix prescription	+0.20
Elevated BNP	+0.15
Historical HCC	+0.11
Cardiology visit	+0.08

SHAP values allow coders and compliance teams to understand why the model flagged an encounter.

12. Drift Monitoring

- Population Stability Index (PSI)
- Kolmogorov – Smirnov (K-S) test
- Feature importance drift
- Coding policy change monitoring

PSI Range	Interpretation
0.00 – 0.10	Stable
0.10 – 0.25	Moderate drift
≥ 0.25	Significant drift

13. Financial Impact Estimation

Sample Opportunity Portfolio

HCC Type	Opportunities	Avg RAF Impact
CHF	12,500	0.32
CKD	18,200	0.24
COPD	9,800	0.18
Diabetes	7,400	0.15

Estimated RAF Opportunity = Σ (Opportunities $\times$ Avg RAF Impact)

Finance teams convert RAF opportunity into projected revenue using CMS payment factors.

14. AI-Assisted Coding Summary

Input to LLM

- Predicted Probability: 0.92
- SHAP Drivers
- Clinical Note Snippets
- Lab Summary
- Medication Summary

Generated Summary

Evidence suggests a potential undocumented Congestive Heart Failure diagnosis. The member has a prior CHF history, is currently prescribed loop diuretics, has elevated BNP levels, and the cardiology note references reduced ejection fraction. No corresponding ICD-10 diagnosis was submitted on the current claim. Recommend coder review for possible HCC capture.

The AI layer provides a coder-ready explanation but does not make the final coding decision.

15. MLflow Logging & Governance

- Model name and version
- Training date
- Feature list hash
- Hyperparameters
- AUROC / Precision / Recall
- SHAP artifacts
- Drift reports
- Scoring output files
- Approval workflow status

Production Governance

All scoring runs are versioned in MLflow. Model promotion requires validation by data science, coding operations, compliance, and actuarial teams. Human coders remain responsible for all final diagnosis submission decisions.

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